

Training Configuration and Summary:

Sequencing:

- timesteps: 30
- batch_size: 128

Model Hyperparameters:

- min_units: 50
- max_units: 150
- units_step: 50
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh']
- optimizers: ['adam']

Tuning:

- max_epochs: 100
- hyperband_iterations: 3

LSTM Multi-Step Training Report

Report Generated: 2025-04-30 14:15:01

Training Start: 2025-04-23_17-09-41

Training End: 2025-04-23_17-40-17

Training Duration: 30.60 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: adam

Dropout Rate: 0.20

MAE: 1.0780

MSE: 1.9251

RMSE: 1.3875

R2 Score: 0.6818

Multi-Step Forecast:

Forecast Steps: 10

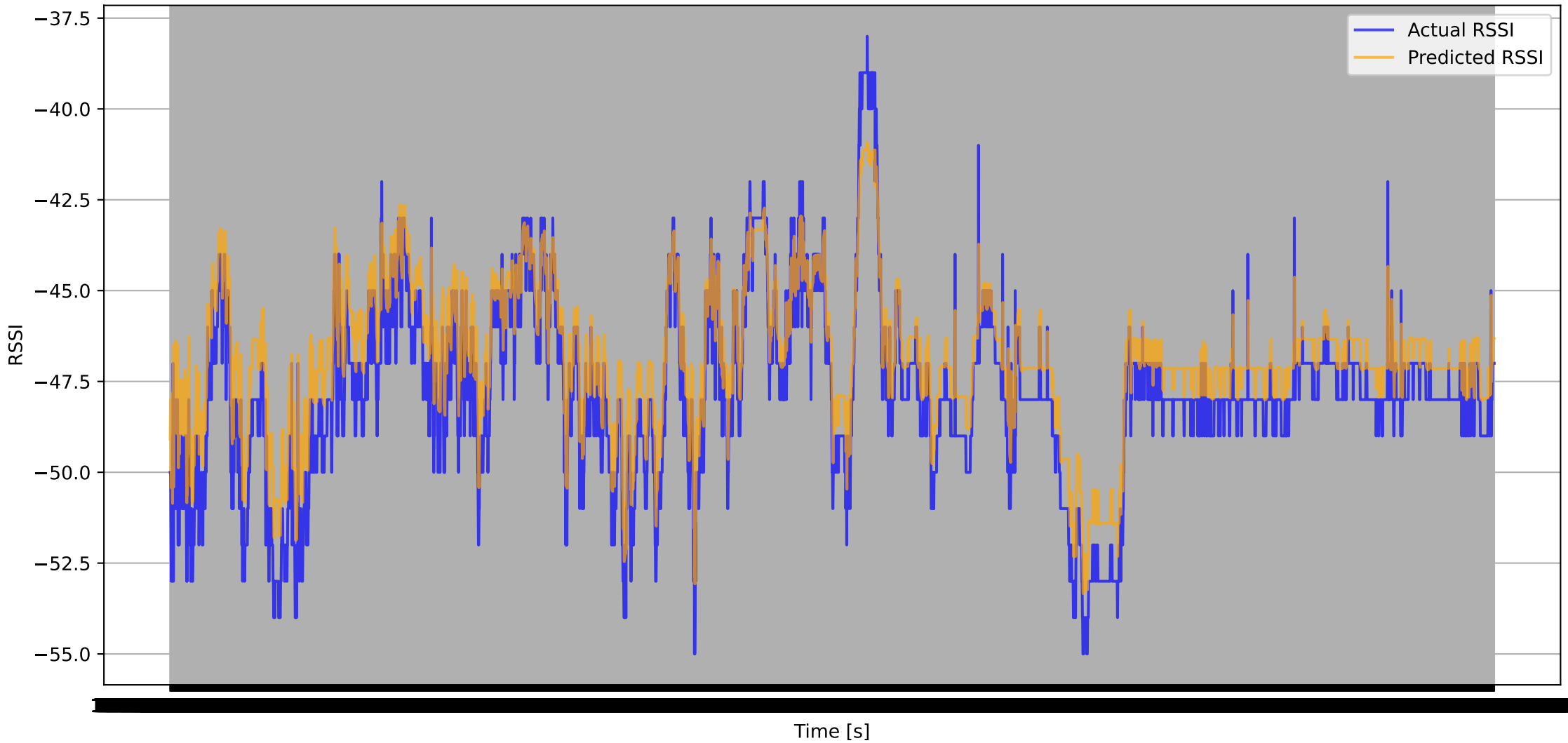
MAE: 0.0376

MSE: 0.0019

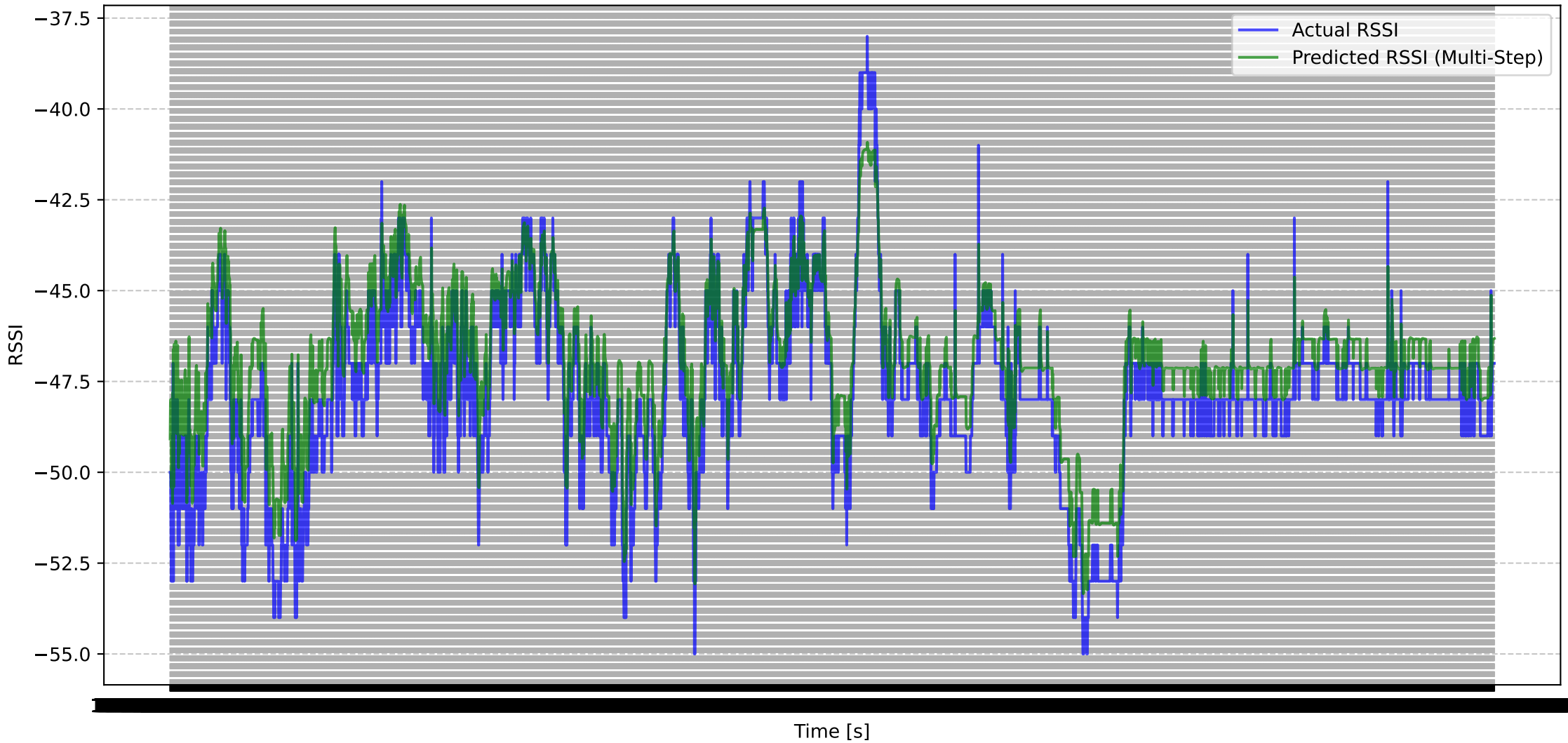
RMSE: 0.0438

R2 Score: 0.7340

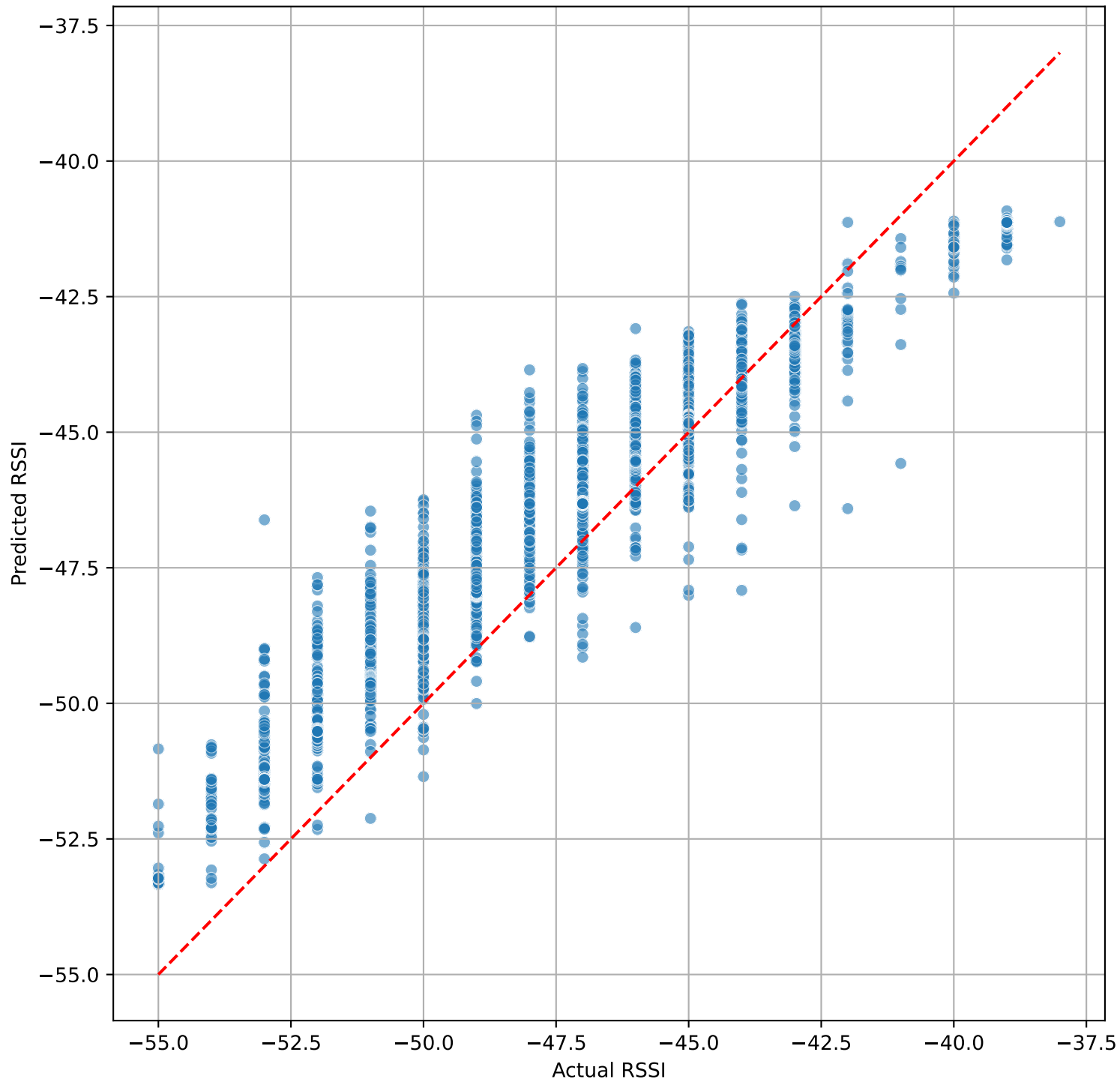
Actual vs Predicted RSSI (First Step)



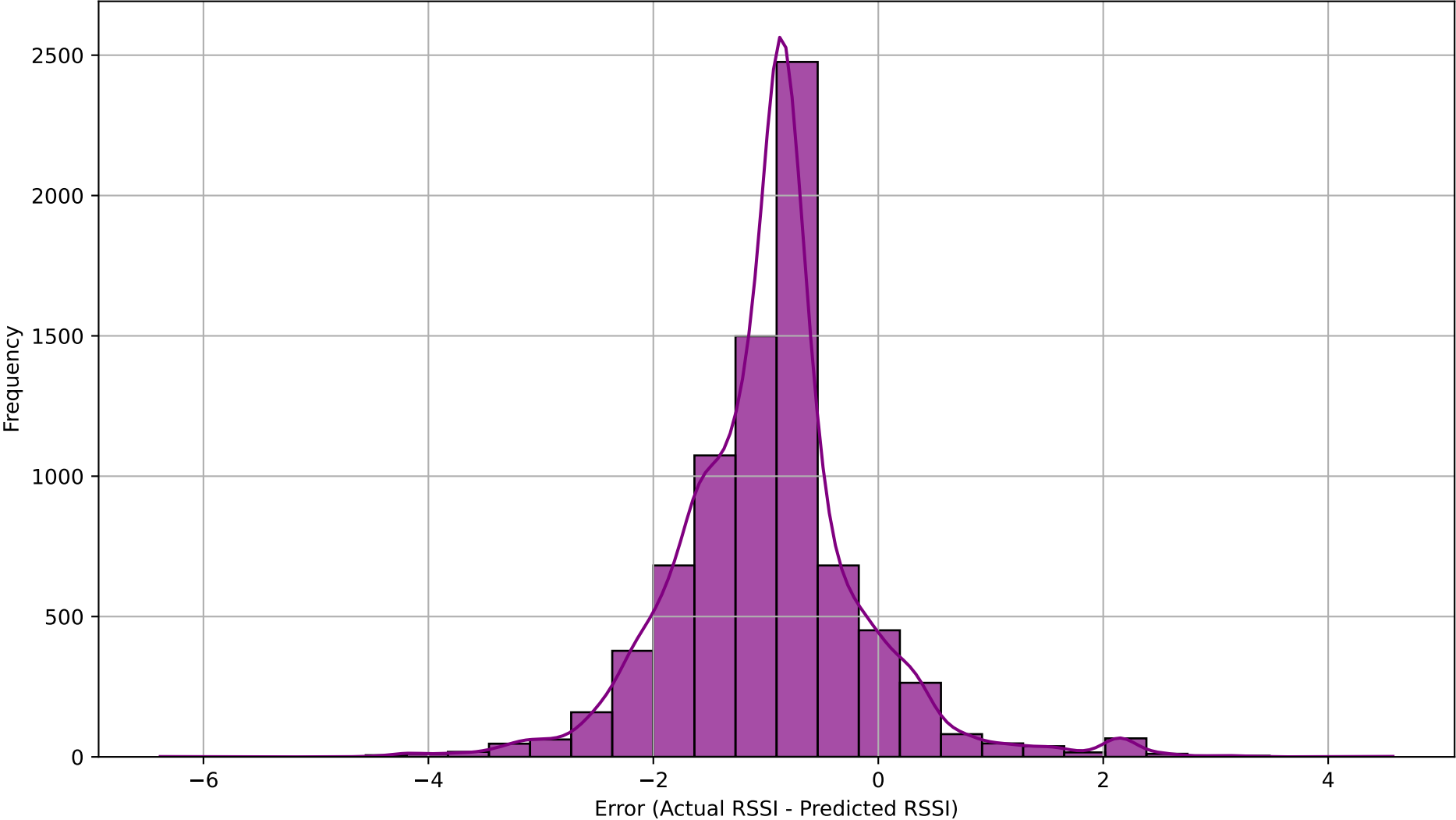
Actual vs Predicted RSSI (First Step vs Multi-Step)



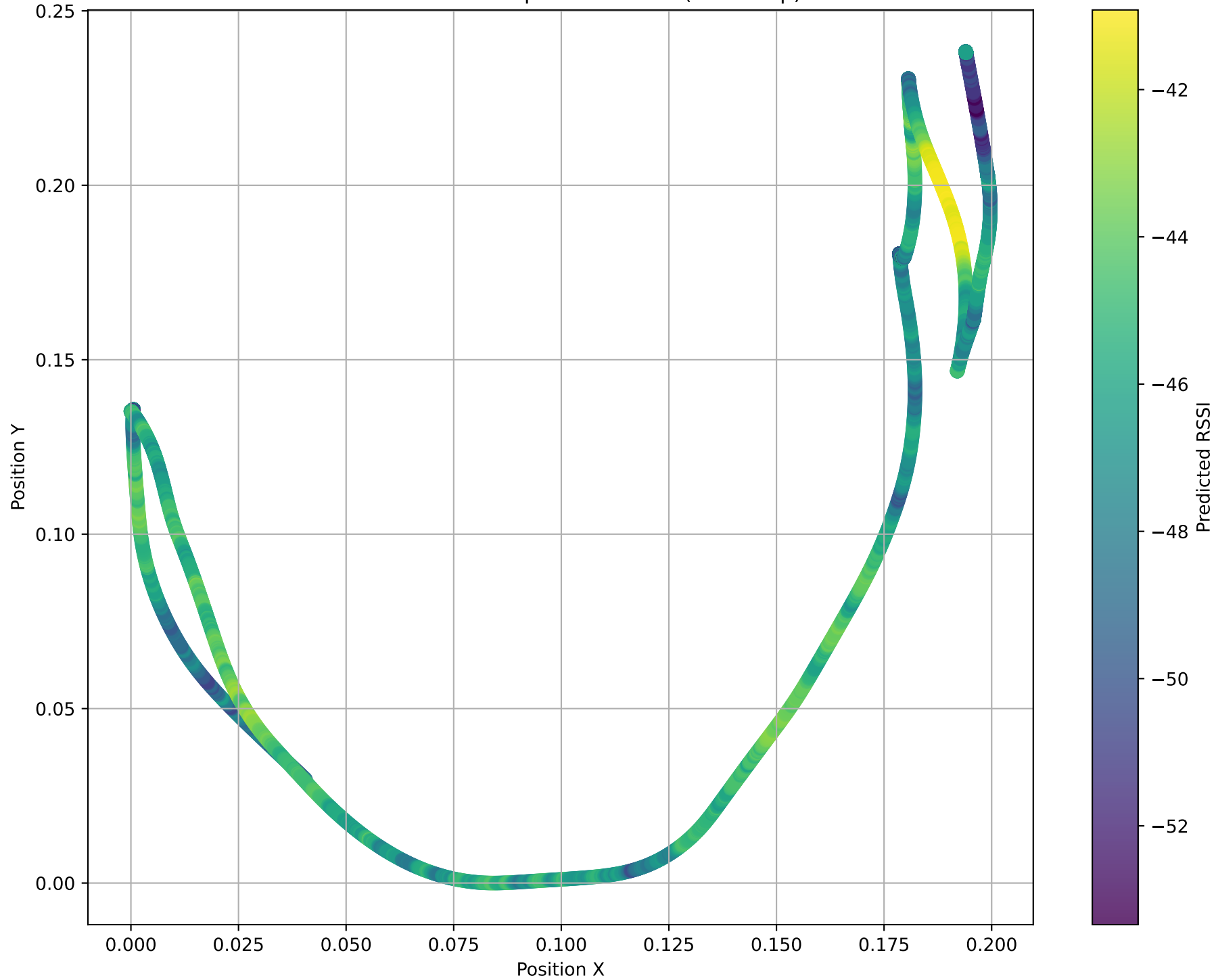
Actual vs Predicted RSSI (First Step)



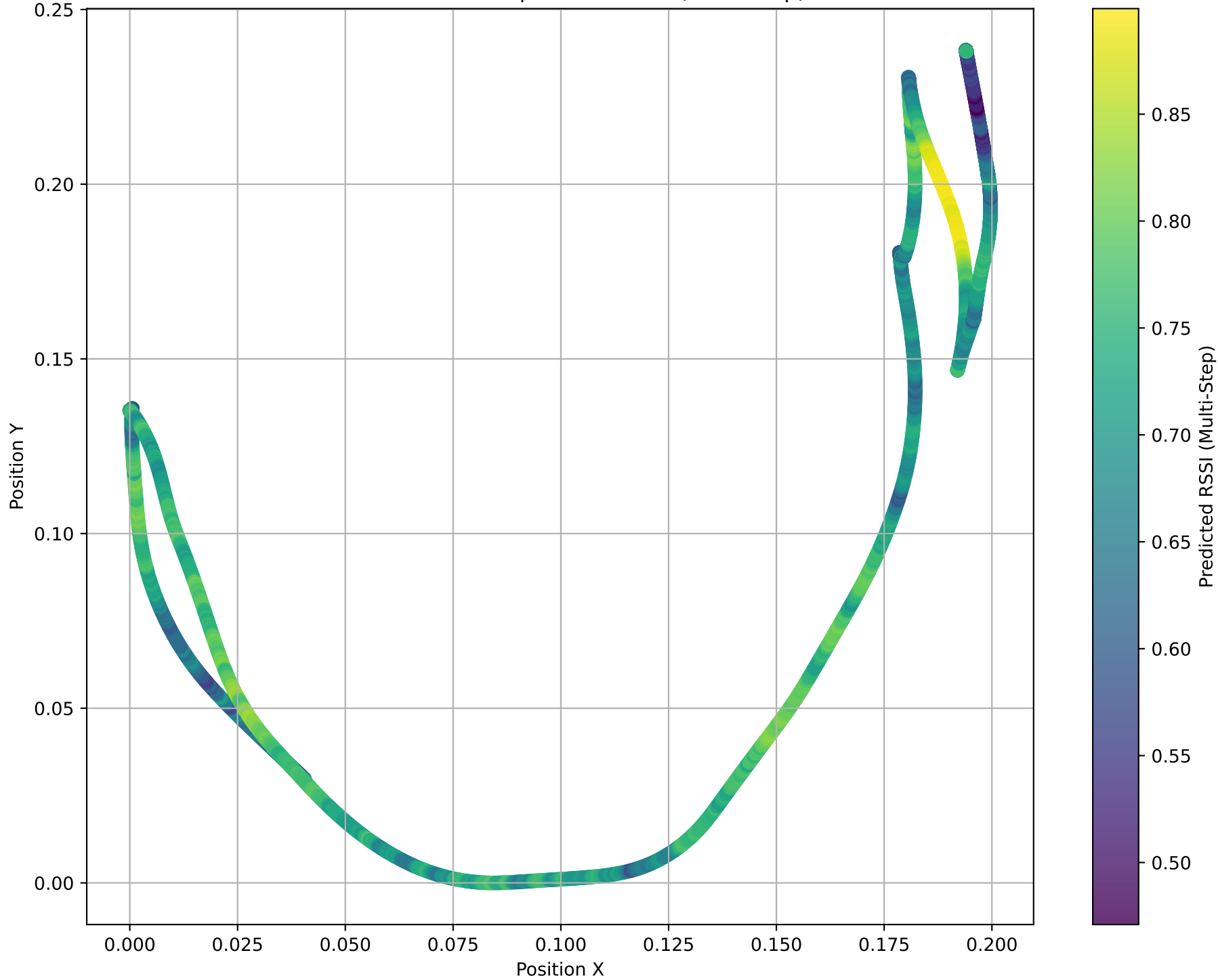
Distribution of Prediction Errors (First Step)



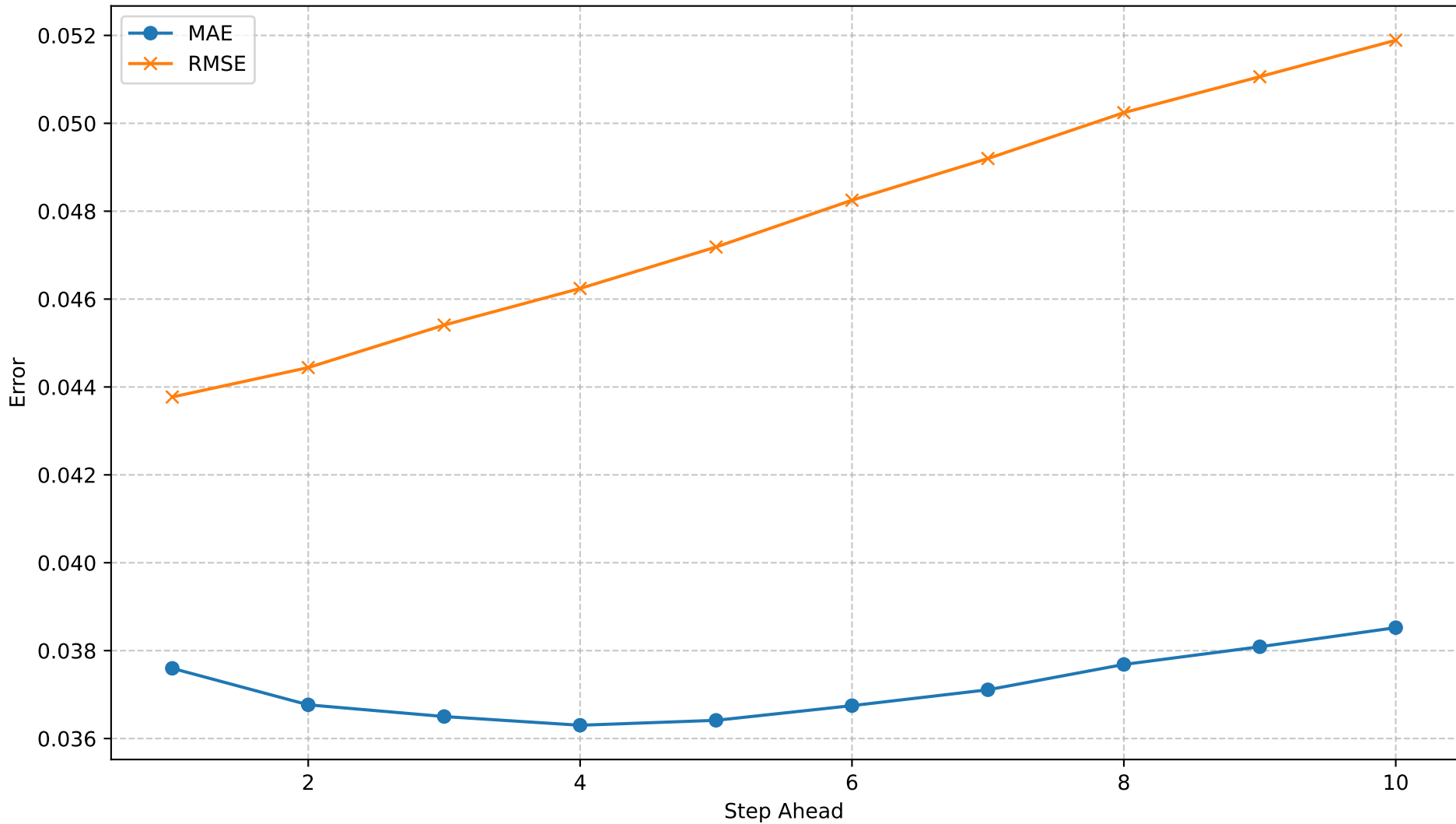
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step)



Multi-Step Forecast Error by Step



Resumo Automático:

- Forecast Steps configurados: 10
- Passos com MAE $\leq$ 2.0: 10 passo(s)
- MAE no último passo: 0.0385

[V] O modelo mantém boa acurácia durante quase toda a sequência de previsões.